

Analyzing the Data

Replication File II for the TEPS School Distinction Paper

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2026-10-02

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1 Setup

```
## Data packages
library(tidyverse)
library(haven)
library(fastDummies)

## Model packages
library(fixest)
library(nnet)
library(mlogit)

## Stat test packages
library(sandwich)
library(lmtest)

## Display packages
library(broom)
library(modelsummary)
library(kableExtra)
```

```
## file_stamp(): provenance stamp for input files
file_stamp <- function(...) {
  paths <- c(...)
  fi <- file.info(paths)
  data.frame(
    file      = basename(paths),
    size_mb   = round(fi$size / 1024^2, 1),
    modified  = format(fi$mtime, "%Y-%m-%d"),
    md5       = unname(tools::md5sum(paths)),
    row.names = NULL
  )
}
```

```
## ms_cluster(): multinom estimates with cluster-robust SEs, packaged for modelsummary
##   (modelsummary(vcov = ...) silently keeps the iid SEs for multinom models,
##   because parameters::standard_error.multinom() ignores the vcov argument)
ms_cluster <- function(model, cluster) {
  V <- vcovCL(model, cluster = cluster) # HCO with G/(G-1) adjustment
  b <- coef(model)                     # rows = response levels, cols = terms
  tidy <- tibble(
```

```

    term      = rep(colnames(b), times = nrow(b)),
    response = rep(rownames(b), each = ncol(b)),
    estimate = as.vector(t(b))
  ) %>%
  mutate(std.error = unname(sqrt(diag(V))[paste(response, term, sep = ":")]),
         statistic = estimate / std.error,
         p.value    = 2 * pnorm(-abs(statistic)),
         conf.low   = estimate - qnorm(0.975) * std.error,
         conf.high  = estimate + qnorm(0.975) * std.error)
  stopifnot(!anyNA(tidy$std.error))
  glance <- get_gof(model)
  glance$vcov.type <- paste("by:", paste(all.vars(cluster), collapse = " + "))
  structure(list(tidy = tidy, glance = glance), class = "modelsummary_list")
}

```

```

## Stamp data
file_stamp("data/TEPS-Cleaned.dta")

```

file	size_mb	modified	md5
TEPS-Cleaned.dta	6.7	2026-10-02	fb98e0d36e2c5e5616b339120303ca8a

```

## Read data
teps <- read_dta("data/TEPS-Cleaned.dta")
#glimpse(teps)

## Factorize and rename variables
teps$geoarea <- teps$geoarea %>% as.factor()
levels(teps$geoarea) <- c("Taipei", "Taoyuan", "Taichung", "Tainan", "Kaohsiung", "Taitung")

teps$schtype <- as.factor(teps$schtype)
levels(teps$schtype) <- c("ordi", "priv", "star")

teps$pses <- as.factor(teps$pses)
levels(teps$pses) <- c("E0I0", "E0I1", "E1I0", "E1I1")
teps$pses <- relevel(teps$pses, ref = "E0I1")

teps$pedu <- as.factor(teps$pedu)
teps$income <- as.factor(teps$income)
teps <- teps %>% rename(pedu_bi = pedu2, income_bi = income2)

```

```
teps$pethn <- as.factor(teps$pethn)
teps$pocc <- as.factor(teps$pocc)
```

```
## Check school type distribution
```

```
teps %>% group_by(sch_id) %>%
  summarise(sch_id = first(sch_id), schtype = first(schtype), geoarea = first(geoarea)) %>%
  xtabs(~ schtype + geoarea, .) %>%
  kable()
```

	Taipei	Taoyuan	Taichung	Tainan	Kaohsiung	Taitung
ordi	78	24	52	23	29	18
priv	18	7	9	14	6	0
star	19	6	13	6	7	4

```
## Score Distribution
```

```
teps %>% group_by(schtype) %>% summarise(
  n = n(),
  missing = sum(is.na(cogability)),
  cog_avg = mean(cogability, na.rm = TRUE),
  cog_std = sd(cogability, na.rm = TRUE),
  cog_Q1 = quantile(cogability, probs = c(0.25), na.rm = TRUE),
  cog_Q2 = median(cogability, na.rm = TRUE),
  cog_Q3 = quantile(cogability, probs = c(0.75), na.rm = TRUE)) %>%
  mutate_if(is.numeric, round, digits=3)
```

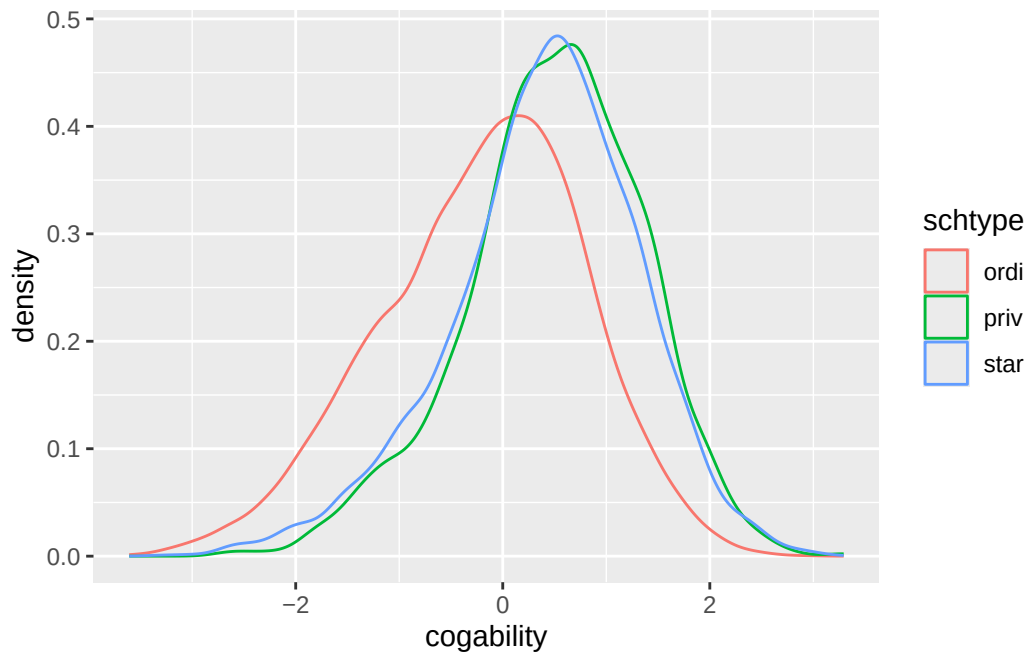
schtype	n	missing	cog_avg	cog_std	cog_Q1	cog_Q2	cog_Q3
ordi	13071	28	-0.201	0.995	-0.857	-0.112	0.503
priv	2306	4	0.498	0.860	-0.008	0.549	1.098
star	3676	5	0.397	0.917	-0.132	0.472	1.018

```
teps %>% group_by(schtype) %>% summarise(
  n = n(),
  missing = sum(is.na(finscore)),
  score_avg = mean(finscore, na.rm = TRUE),
  score_std = sd(finscore, na.rm = TRUE),
  score_Q1 = quantile(finscore, probs = c(0.25), na.rm = TRUE),
  score_Q2 = median(finscore, na.rm = TRUE),
  score_Q3 = quantile(finscore, probs = c(0.75), na.rm = TRUE)) %>%
  mutate_if(is.numeric, round, digits=1)
```

schtype	n	missing	score_avg	score_std	score_Q1	score_Q2	score_Q3
ordi	13071	975	144.8	70.9	82	141	205
priv	2306	558	193.7	59.0	153	204	240
star	3676	264	190.6	65.5	142	205	244

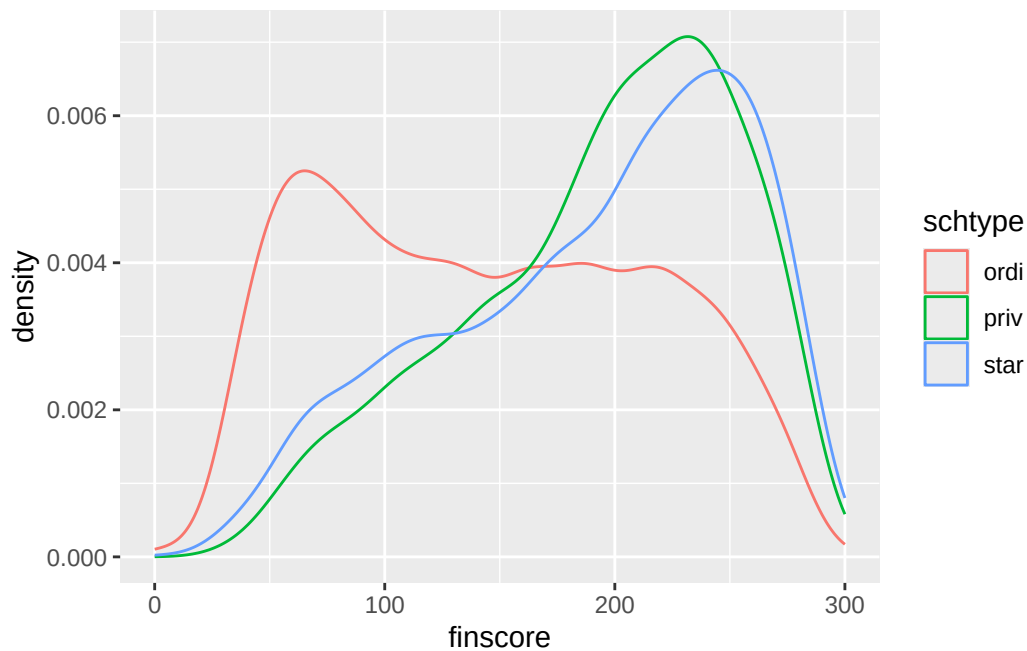
```
## Visualization
teps %>% ggplot(aes(x = cogability, color = schtype)) +
  labs(x = "cogability") + geom_density()
```

Warning: Removed 37 rows containing non-finite outside the scale range (`stat\_density()`).



```
teps %>% ggplot(aes(x = finscore, color = schtype)) +
  labs(x = "finscore") + geom_density()
```

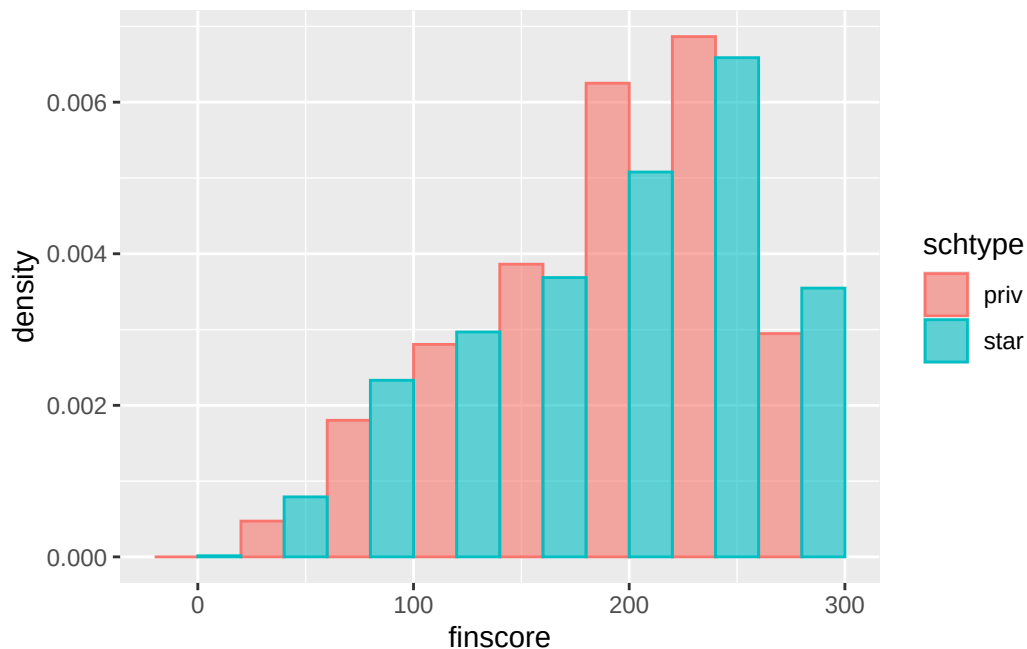
Warning: Removed 1797 rows containing non-finite outside the scale range (`stat\_density()`).



```
#steps %>% filter(geoarea == "Taipei") %>% ggplot(aes(x = finscore, color = schtype)) + geom_density()
#steps %>% filter(geoarea == "Taichung") %>% ggplot(aes(x = finscore, color = schtype)) + geom_density()

steps %>% filter(schtype != "ordi") %>%
  ggplot(aes(x = finscore, color = schtype, fill = schtype, after_stat(density))) +
  labs(x = "finscore") +
  geom_histogram(position = "dodge", binwidth = 40, alpha = 0.6)
```

Warning: Removed 822 rows containing non-finite outside the scale range (``stat_bin()``).



```
## Select variables
teps <- teps %>%
  select(stud_id, sch_id, geoarea, priv, star, schtype,
         male, onlychild, edstchild, ygstchild, cogability, marriage, pethn,
         pedu, pedu_bi, income, income_bi, pses, pocc, pboss, pmgm,
         pexpedu, book, art, reloc,
         acadcomm, talentcls, abstudy, edusav)

## Drop NA
teps <- teps %>% drop_na() # n = 19053 -> 18465
#sapply(teps, function(y) sum(is.na(y))) %>% data.frame(na_count = .)

## Variable Distributions
teps %>% count(geoarea) %>% mutate(pct = round(n * 100 / sum(n), 2))
```

geoarea	n	pct
Taipei	6358	34.43
Taoyuan	1767	9.57
Taichung	4752	25.74

geoarea	n	pct
Tainan	2183	11.82
Kaohsiung	2309	12.50
Taitung	1096	5.94

```
teps %>% count(pedu) %>% mutate(pct = round(n * 100 / sum(n), 2))
```

pedu	n	pct
1	4953	26.82
2	7896	42.76
3	3111	16.85
4	1857	10.06
5	648	3.51

```
teps %>% count(income) %>% mutate(pct = round(n * 100 / sum(n), 2))
```

income	n	pct
1	1824	9.88
2	7589	41.10
3	6500	35.20
4	1652	8.95
5	900	4.87

```
teps %>% count(pses) %>% mutate(pct = round(n * 100 / sum(n), 2))
```

pses	n	pct
E0I1	1350	7.31
E0I0	14610	79.12
E1I0	1303	7.06
E1I1	1202	6.51

2 Description

```
## Make dummies for factors
teps2 <- teps %>% dummy_cols(select_columns = c("pethn", "pses", "pocc")) %>%
  select(-stud_id, -sch_id, -geoarea, -schtype, -pethn, -pedu, -income, -pses, -pocc)

## Calculate means
vars <- teps2 %>% colnames()
sum1 <- teps2 %>% summarise_all(mean)
sum2 <- teps2 %>% group_by(priv, star) %>% summarise_all(mean)

## Create a data frame
sumstat <- data.frame(
  Variable = vars,
  Mean_All = as.numeric(sum1),
  Ordinary = as.numeric(sum2[1,]),
  Private = as.numeric(sum2[3,]),
  Star = as.numeric(sum2[2,]))

# Output
sumstat <- sumstat %>% mutate_if(is.numeric, round, digits=3)
sumstat
```

Variable	Mean_All	Ordinary	Private	Star
priv	0.122	0.000	1.000	0.000
star	0.194	0.000	0.000	1.000
male	0.513	0.512	0.550	0.496
onlychild	0.058	0.051	0.083	0.066
edstchild	0.345	0.329	0.397	0.369
ygstchild	0.367	0.364	0.344	0.390
cogability	0.008	-0.191	0.501	0.400
marriage	0.906	0.892	0.944	0.928
pedu_bi	0.136	0.080	0.230	0.273
income_bi	0.138	0.083	0.294	0.234
pboss	0.145	0.110	0.295	0.175
pmgm	0.178	0.138	0.272	0.259
pexpedu	0.478	0.397	0.685	0.632

Variable	Mean_All	Ordinary	Private	Star
book	0.354	0.319	0.429	0.430
art	0.129	0.108	0.150	0.186
reloc	0.324	0.270	0.404	0.463
acadcomm	0.256	0.239	0.339	0.264
talentcls	0.657	0.591	0.830	0.781
abstudy	0.066	0.043	0.153	0.092
edusav	0.636	0.592	0.798	0.689
pethn_1	0.733	0.741	0.760	0.689
pethn_2	0.056	0.063	0.034	0.046
pethn_3	0.042	0.033	0.054	0.067
pethn_4	0.136	0.120	0.143	0.190
pethn_5	0.032	0.043	0.009	0.008
pses_E0I1	0.073	0.052	0.166	0.089
pses_E0I0	0.791	0.868	0.604	0.638
pses_E1I0	0.071	0.049	0.102	0.127
pses_E1I1	0.065	0.031	0.128	0.145
pocc_1	0.200	0.242	0.106	0.112
pocc_2	0.243	0.236	0.257	0.258
pocc_3	0.113	0.088	0.162	0.172
pocc_4	0.223	0.187	0.305	0.299
pocc_5	0.221	0.247	0.171	0.160

```
# write_csv(sumstat, "export/summary_statistics.csv")
```

3 Private Schools

```
## M1: Independent
Priv1 <- feglm(
  priv ~ pedu_bi + income_bi,
  family = "binomial",
  data = teps,
  cluster = ~sch_id
)

## M2: Independent + Control
Priv2 <- feglm(
  priv ~ pedu_bi + income_bi +
    male + onlychild + edstchild + ygstchild + cogability + marriage +
    pethn + geoarea + pocc + pboss + pmgm,
  family = "binomial",
  data = teps,
  cluster = ~sch_id
)

## M3: Interaction
teps$peduXincome <- teps$pedu_bi*teps$income_bi

Priv3 <- feglm(
  priv ~ pedu_bi + income_bi + peduXincome +
    male + onlychild + edstchild + ygstchild + cogability + marriage +
    pethn + geoarea + pocc + pboss + pmgm,
  family = "binomial",
  data = teps,
  cluster = ~sch_id
)

## M4: Categorization
Priv4 <- feglm(
  priv ~ pses +
    male + onlychild + edstchild + ygstchild + cogability + marriage +
    pethn + geoarea + pocc + pboss + pmgm,
  family = "binomial",
```

```

data = teps,
cluster = ~sch_id
)

## Show results
modelsummary(list(Priv1, Priv2, Priv3, Priv4),
  coef_map = c(
    "pedu_bi", "income_bi", "peduXincome",
    "psesE0I0", "psesE1I0", "psesE1I1",
    "male", "onlychild", "edstchild", "ygstchild", "cogability",
    "marriage", "pboss", "pmgm",
    "(Intercept)"),
  gof_omit = "R2|BIC|RMSE",
  title = "Private vs. Public",
  stars = TRUE,
  output = "markdown")

```

Table 1: Private vs. Public

	(1)	(2)	(3)	(4)
pedu_bi	0.313* (0.126)	-0.083 (0.113)	0.207+ (0.113)	
income_bi	1.026*** (0.111)	0.628*** (0.092)	0.857*** (0.093)	
peduXincome			-0.671*** (0.141)	
psesE0I0				-0.857*** (0.093)
psesE1I0				-0.650*** (0.124)
psesE1I1				-0.463** (0.151)
male		0.204 (0.151)	0.207 (0.151)	0.207 (0.151)
onlychild		0.644*** (0.111)	0.637*** (0.111)	0.637*** (0.111)
edstchild		0.203** (0.078)	0.199* (0.078)	0.199* (0.078)
ygstchild		0.029 (0.075)	0.031 (0.075)	0.031 (0.075)

	(1)	(2)	(3)	(4)
cogability		0.462*** (0.077)	0.465*** (0.078)	0.465*** (0.078)
marriage		0.353** (0.132)	0.353** (0.132)	0.353** (0.132)
pboss		0.940*** (0.087)	0.904*** (0.087)	0.904*** (0.087)
pmgm		0.400*** (0.081)	0.395*** (0.082)	0.395*** (0.082)
(Intercept)	-2.225*** (0.161)	-3.791*** (0.363)	-3.809*** (0.363)	-2.952*** (0.366)
Num.Obs.	18465	18465	18465	18465
AIC	13261.3	11597.1	11572.2	11572.2
• p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001				

```
# modelsummary(list(Priv1, Priv2, Priv3, Priv4),
#               stars = TRUE, output = "export/private.xlsx")
```

4 Star Schools

```
## Multinom
Star1 <- teps %>%
  multinom(schtype ~ pedu_bi + income_bi + peduXincome +
            male + onlychild + edstchild + ygstchild + cogability + marriage +
            pethn + geoarea + pocc + pboss + pmgm,
            data = .)
```

```
# weights:  78 (50 variable)
initial  value 20285.875910
iter   10 value 14512.162506
iter   20 value 13802.242081
iter   30 value 13447.088303
iter   40 value 13310.652599
iter   50 value 13256.967502
final   value 13250.752291
converged
```

```
Star1_cl <- ms_cluster(Star1, cluster = ~ sch_id)
```

```
## Multinom with a different reference group
teps$schtype <- relevel(teps$schtype, ref = "priv")
Star2 <- teps %>%
  multinom(schtype ~ pedu_bi + income_bi + peduXincome +
            male + onlychild + edstchild + ygstchild + cogability + marriage +
            pethn + geoarea + pocc + pboss + pmgm,
            data = .)
```

```
# weights:  78 (50 variable)
initial  value 20285.875910
iter   10 value 14255.294740
iter   20 value 13957.533234
iter   30 value 13495.596695
iter   40 value 13318.764061
iter   50 value 13266.800868
iter   60 value 13251.024942
iter   70 value 13250.752567
final   value 13250.752350
converged
```

```

Star2_cl <- ms_cluster(Star2, cluster = ~ sch_id)

## Results (SEs clustered by school)
modelsummary(list(Star1_cl, Star2_cl),
  shape = term ~ model + response,
  coef_map = c(
    "pedu_bi", "income_bi", "peduXincome",
    "male", "onlychild", "edstchild", "ygstchild", "cogability",
    "marriage", "pboss", "pmgm",
    "(Intercept)"),
  gof_omit = "R2|BIC|RMSE",
  title = "Private & Star vs. Public",
  stars = TRUE,
  output = "markdown")

```

Table 2: Private & Star vs. Public

	(1)		(2)	
	star	priv	star	ordi
pedu_bi	0.779*** (0.087)	0.502*** (0.114)	0.277* (0.121)	-0.502*** (0.114)
income_bi	0.562*** (0.084)	1.030*** (0.095)	-0.468*** (0.109)	-1.030*** (0.095)
peduXincome	-0.194 (0.122)	-0.613*** (0.152)	0.418** (0.148)	0.613*** (0.152)
male	-0.031 (0.070)	0.198 (0.148)	-0.229 (0.164)	-0.198 (0.148)
onlychild	0.308** (0.103)	0.727*** (0.118)	-0.418** (0.128)	-0.727*** (0.118)
edstchild	0.172** (0.064)	0.243** (0.080)	-0.071 (0.090)	-0.243** (0.080)
ygstchild	0.239*** (0.054)	0.097 (0.079)	0.142+ (0.080)	-0.097 (0.079)
cogability	0.481*** (0.037)	0.609*** (0.081)	-0.129 (0.082)	-0.609*** (0.081)
marriage	0.129+ (0.077)	0.378** (0.135)	-0.248+ (0.143)	-0.378** (0.135)
pboss	0.386*** (0.068)	1.023*** (0.090)	-0.636*** (0.097)	-1.023*** (0.090)
pmgm	0.300***	0.491***	-0.191*	-0.491***

	(1)		(2)	
	star	priv	star	ordi
	(0.054)	(0.083)	(0.087)	(0.083)
(Intercept)	-2.813***	-3.952***	1.139**	3.951***
	(0.284)	(0.378)	(0.435)	(0.378)
Num.Obs.	18465		18465	
AIC	26601.5		26601.5	
Std.Errors	by: sch_id		by: sch_id	
• p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001				

```
# modelsummary(list(Star1_cl, Star2_cl),
#                 shape = term ~ model + response,
#                 stars = TRUE, output = "export/star.xlsx")
```

4.1 Check

```
## Cross-check: mlogit + vcovCL reproduces the clustered SEs in Star1_cl
teps_choice <- mlogit.data(
  teps,
  choice = "schtype",
  shape = "wide",
  chid.var = "stud_id")

# mlogit syntax:
# choice ~ alternative_specific_variables | individual_specific_variables
Star1_ml <- mlogit(
  schtype ~ 0 |
  pedu_bi + income_bi + peduXincome +
  male + onlychild + edstchild + ygstchild + cogability + marriage +
  pethn + geoarea + pocc + pboss + pmgm,
  data = teps_choice, reflevel = "ordi"
)

## teps_choice keeps teps' row order, so teps$sch_id lines up with its choice situations
ct <- coeftest(Star1_ml, vcov. = vcovCL(Star1_ml, cluster = teps$sch_id))

Star1_cl$tidy %>%
  filter(term %in% c("pedu_bi", "income_bi", "peduXincome")) %>%
```

```
mutate(se_mlogit = ct[paste(term, response, sep = ":"), "Std. Error"]) %>%
select(response, term, estimate, std.error, se_mlogit)
```

response	term	estimate	std.error	se_mlogit
priv	pedu_bi	0.5015846	0.1143144	0.1143114
priv	income_bi	1.0303464	0.0950642	0.0950624
priv	peduXincome	-0.6130382	0.1521158	0.1521107
star	pedu_bi	0.7785386	0.0865816	0.0865834
star	income_bi	0.5618983	0.0843543	0.0843563
star	peduXincome	-0.1944697	0.1222209	0.1222233

4.2 Sequence

```
## Tidy coefficient tests
coeftest_tidy = function(tested) {
  tested %>%
    tidy() %>%
    filter(str_starts(term, "pscs")) %>%
    mutate(
      p.sig = case_when(
        p.value < .001 ~ "***",
        p.value < .01 ~ "**",
        p.value < .05 ~ "*",
        p.value < .1 ~ ".",
        .default = ""
      ),
      estimate = round(estimate, 3),
      std.error = round(std.error, 3),
      statistic = round(statistic, 2)
    )
}

## A. Private vs. Public (SES Cat)
teps$pscs <- relevel(teps$pscs, ref = "E0I0")
PrivSES <- feglm(
  priv ~ pscs +
    male + onlychild + edstchild + ygstchild + cogability + marriage +
    pethn + geoarea + pocc + pboss + pmgm,
  family = "binomial",
```

```

data = teps,
cluster = ~sch_id
)
PrivSES %>% coeftest() %>% coeftest_tidy()

```

term	estimate	std.error	statistic	p.value	p.sig
psesE0I1	0.857	0.093	9.23	0.0000000	***
psesE1I0	0.207	0.113	1.84	0.0657286	.
psesE1I1	0.394	0.171	2.30	0.0213947	*

```

## B. Multinom (SES Cat)
teps$schtype <- relevel(teps$schtype, ref = "ordi")
teps_choice <- mlogit.data(
  teps,
  choice = "schtype",
  shape = "wide",
  chid.var = "stud_id")

StarSES <- mlogit(
  schtype ~ 0 |
  pses +
  male + onlychild + edstchild + ygstchild + cogability + marriage +
  pethn + geoarea + pocc + pboss + pmgm,
  data = teps_choice, reflevel = "ordi"
)

StarSES %>%
  coeftest(vcov. = vcovCL(StarSES, cluster = teps$sch_id)) %>%
  coeftest_tidy() %>%
  arrange(str_detect(term, ":star"), term)

```

term	estimate	std.error	statistic	p.value	p.sig
psesE0I1:priv	1.030	0.095	10.84	0.00e+00	***
psesE1I0:priv	0.502	0.114	4.39	1.15e-05	***
psesE1I1:priv	0.919	0.170	5.39	1.00e-07	***
psesE0I1:star	0.562	0.084	6.66	0.00e+00	***
psesE1I0:star	0.779	0.087	8.99	0.00e+00	***
psesE1I1:star	1.146	0.132	8.69	0.00e+00	***

```
## C. Sequence 1 (Priv & Star vs. Ordi) (SES Cat)
teps <- teps %>% mutate(privstar = ifelse(schtype == "ordi", 0, 1))
Seq1SES <- feglm(
  privstar ~ pses +
    male + onlychild + edstchild + ygstchild + cogability + marriage +
    pethn + geoarea + pocc + pboss + pmgm,
  family = "binomial",
  data = teps,
  cluster = ~sch_id
)
Seq1SES %>% coeftest() %>% coeftest_tidy()
```

term	estimate	std.error	statistic	p.value	p.sig
psesE0I1	0.770	0.074	10.35	0	***
psesE1I0	0.685	0.080	8.56	0	***
psesE1I1	1.062	0.119	8.91	0	***

```
## C. Sequence 2 (Star vs. Priv; Ordi excluded) (SES Cat)
Seq2SES <- feglm(
  star ~ pses +
    male + onlychild + edstchild + ygstchild + cogability + marriage +
    pethn + geoarea + pocc + pboss + pmgm,
  family = "binomial",
  data = teps %>% filter(schtype != "ordi"),
  cluster = ~sch_id
)
Seq2SES %>% coeftest() %>% coeftest_tidy()
```

term	estimate	std.error	statistic	p.value	p.sig
psesE0I1	-0.504	0.113	-4.47	0.0000080	***
psesE1I0	0.258	0.122	2.12	0.0338271	*
psesE1I1	0.239	0.184	1.30	0.1938582	

```
## Again, with E0I1 as ref
teps$pses <- relevel(teps$pses, ref = "E0I1")
Seq2SES_inc <- feglm(
  star ~ pses +
    male + onlychild + edstchild + ygstchild + cogability + marriage +
```

```

    pethn + geoarea + pocc + pboss + pmgm,
    family = "binomial",
    data = teps %>% filter(schtype != "ordi"),
    cluster = ~sch_id,
  )
Seq2SES_inc %>% coeftest() %>% coeftest_tidy()

```

term	estimate	std.error	statistic	p.value	p.sig
psese0I0	0.504	0.113	4.47	8.0e-06	***
psese1I0	0.762	0.140	5.45	1.0e-07	***
psese1I1	0.743	0.161	4.63	3.8e-06	***

```

## Again, with E1I0 as ref
teps$psese <- releve(teps$psese, ref = "E1I0")
Seq2SES_edu <- update(Seq2SES_inc, data = teps %>% filter(schtype != "ordi"))
Seq2SES_edu %>% coeftest() %>% coeftest_tidy()

```

term	estimate	std.error	statistic	p.value	p.sig
psese0I1	-0.762	0.140	-5.45	0.0000001	***
psese0I0	-0.258	0.122	-2.12	0.0338271	*
psese1I1	-0.019	0.150	-0.13	0.8974910	

```

## Show sample size
teps %>% filter(schtype != "ordi") %>% nrow()

```

```
[1] 5842
```

5 Parenting

```
# Define dependent variables for the models (mediators)
dep_vars <- c("pexpedu", "book", "art", "reloc",
              "acadcomm", "talentcls", "abstudy", "edusav")

# Loop through each dependent variable and save models separately
for (i in seq_along(dep_vars)) {
  formula <- as.formula(paste(dep_vars[i], "~ pedu_bi + income_bi + peduXincome +
                              male + onlychild + edstchild + ygstchild +
                              marriage + pethn + geoarea +
                              pocc + pboss + pmgm"))
  model <- feglm(formula, family = "binomial", cluster = ~sch_id, data = teps)
  assign(paste0("Prt", i), model)
}

## Results
modelsummary(
  list("pexpedu" = Prt1, "book" = Prt2, "art" = Prt3, "reloc" = Prt4),
  coef_map = c("pedu_bi", "income_bi", "peduXincome"),
  gof_omit = "R2|BIC|RMSE",
  title = "Intensive Parenting",
  stars = TRUE,
  output = "markdown"
)
```

Table 3: Intensive Parenting

	pexpedu	book	art	reloc
pedu_bi	1.154*** (0.073)	0.475*** (0.064)	0.530*** (0.085)	0.230*** (0.065)
income_bi	0.623*** (0.069)	0.222*** (0.063)	0.242** (0.090)	0.384*** (0.065)
peduXincome	0.015 (0.129)	-0.050 (0.103)	0.003 (0.130)	-0.300** (0.104)
Num.Obs.	18465	18465	18465	18465
AIC	22609.0	23098.9	13601.9	22578.8
• p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001				

```

modelsummary(
  list("acadcomm" = Prt5, "talentcls" = Prt6, "abstudy" = Prt7, "edusav" = Prt8),
  coef_map = c("pedu_bi", "income_bi", "peduXincome"),
  gof_omit = "R2|BIC|RMSE",
  title = "Concerted Outsourcing",
  stars = TRUE,
  output = "markdown"
)

```

Table 4: Concerted Outsourcing

	acadcomm	talentcls	abstudy	edusav
pedu_bi	0.067 (0.071)	0.887*** (0.089)	0.638*** (0.097)	0.152* (0.071)
income_bi	0.299*** (0.069)	0.637*** (0.081)	0.952*** (0.093)	0.841*** (0.078)
peduXincome	-0.355** (0.111)	-0.045 (0.180)	-0.372** (0.140)	-0.315** (0.116)
Num.Obs.	18465	18465	18465	18465
AIC	20725.4	21534.5	7952.3	23160.8

• $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

```

## Multinom
teps$schtype <- releve(teps$schtype, ref = "ordi")
Mech1 <- teps %>%
  multinom(schtype ~ pedu_bi + income_bi + peduXincome +
    pexpedu + book + art + reloc +
    acadcomm + talentcls + abstudy + edusav +
    male + onlychild + edstchild + ygstchild + cogability + marriage +
    pethn + geoarea + pocc + pboss + pmgm,
    data = .)

```

```

# weights: 102 (66 variable)
initial value 20285.875910
iter 10 value 14766.011921
iter 20 value 13987.348502
iter 30 value 13387.023626
iter 40 value 12996.715683
iter 50 value 12945.508139
iter 60 value 12932.044806

```

```
iter 70 value 12927.507102
iter 80 value 12920.908794
final value 12920.833224
converged
```

```
Mech1_cl <- ms_cluster(Mech1, cluster = ~ sch_id)

## Multinom with a different reference group
teps$schtype <- relevel(teps$schtype, ref = "priv")
Mech2 <- teps %>%
  multinom(schtype ~ pedu_bi + income_bi + peduXincome +
    pexpedu + book + art + reloc +
    acadcomm + talentcls + abstudy + edusav +
    male + onlychild + edstchild + ygstchild + cogability + marriage +
    pethn + geoarea + pocc + pboss + pmgm,
    data = .)
```

```
# weights: 102 (66 variable)
initial value 20285.875910
iter 10 value 13805.942289
iter 20 value 13592.829764
iter 30 value 13209.872151
iter 40 value 13074.876275
iter 50 value 12943.846748
iter 60 value 12924.182974
iter 70 value 12921.049590
iter 80 value 12920.842637
final value 12920.833267
converged
```

```
Mech2_cl <- ms_cluster(Mech2, cluster = ~ sch_id)

## Results (SEs clustered by school)
modelsummary(list(Mech1_cl, Mech2_cl),
  shape = term ~ model + response,
  coef_map = c(
    "pedu_bi", "income_bi", "peduXincome",
    "pexpedu", "book", "art", "reloc",
    "acadcomm", "talentcls", "abstudy", "edusav",
    "(Intercept)" ),
  gof_omit = "R2|BIC|RMSE",
```

```

title = "Parenting and School Type (with Cognitive Ability)",
stars = TRUE,
output = "markdown")

```

Table 5: Parenting and School Type (with Cognitive Ability)

	(1)		(2)	
	star	priv	star	ordi
pedu_bi	0.657*** (0.086)	0.331** (0.114)	0.327** (0.118)	-0.331** (0.114)
income_bi	0.443*** (0.085)	0.797*** (0.091)	-0.355*** (0.107)	-0.797*** (0.091)
peduXincome	-0.121 (0.123)	-0.490** (0.154)	0.369* (0.151)	0.490** (0.154)
pexpedu	0.300*** (0.051)	0.463*** (0.060)	-0.162* (0.069)	-0.463*** (0.060)
book	0.054 (0.042)	-0.012 (0.067)	0.065 (0.068)	0.012 (0.067)
art	0.186** (0.072)	-0.153+ (0.085)	0.340*** (0.092)	0.153+ (0.085)
reloc	0.557*** (0.110)	0.130 (0.092)	0.428*** (0.128)	-0.130 (0.092)
acadcomm	-0.065 (0.052)	0.272*** (0.058)	-0.337*** (0.068)	-0.272*** (0.058)
talentcls	0.318*** (0.058)	0.543*** (0.075)	-0.226* (0.088)	-0.543*** (0.075)
abstudy	0.152+ (0.086)	0.662*** (0.109)	-0.510*** (0.111)	-0.662*** (0.109)
edusav	0.026 (0.047)	0.533*** (0.066)	-0.507*** (0.072)	-0.533*** (0.066)
(Intercept)	-3.239*** (0.307)	-4.851*** (0.397)	1.612*** (0.462)	4.851*** (0.397)
Num.Obs.	18465		18465	
AIC	25973.7		25973.7	
Std.Errors	by: sch_id		by: sch_id	
• p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001				

```

## Multinom
teps$schtype <- relevel(teps$schtype, ref = "ordi")
Mech3 <- teps %>%

```

```

multinom(schtype ~ pedu_bi + income_bi + peduXincome +
  pexpedu + book + art + reloc +
  acadcomm + talentcls + abstudy + edusav +
  male + onlychild + edstchild + ygstchild + marriage +
  pethn + geoarea + pocc + pboss + pmgm,
  data = .)

```

```

# weights: 99 (64 variable)
initial value 20285.875910
iter 10 value 15550.952896
iter 20 value 14634.512787
iter 30 value 13599.470481
iter 40 value 13200.084104
iter 50 value 13148.175542
iter 60 value 13142.795115
iter 70 value 13138.144186
iter 80 value 13134.341736
final value 13134.331843
converged

```

```

Mech3_cl <- ms_cluster(Mech3, cluster = ~ sch_id)

```

```

## Multinom with a different reference group
teps$schtype <- relevel(teps$schtype, ref = "priv")
Mech4 <- teps %>%
  multinom(schtype ~ pedu_bi + income_bi + peduXincome +
    pexpedu + book + art + reloc +
    acadcomm + talentcls + abstudy + edusav +
    male + onlychild + edstchild + ygstchild + marriage +
    pethn + geoarea + pocc + pboss + pmgm,
    data = .)

```

```

# weights: 99 (64 variable)
initial value 20285.875910
iter 10 value 13796.292774
iter 20 value 13575.954250
iter 30 value 13365.360650
iter 40 value 13248.941009
iter 50 value 13154.729851
iter 60 value 13141.566526

```

```

iter 70 value 13136.477930
iter 80 value 13134.339043
final value 13134.331869
converged

```

```

Mech4_cl <- ms_cluster(Mech4, cluster = ~ sch_id)

## Results (SEs clustered by school)
modelsummary(list(Mech3_cl, Mech4_cl),
  shape = term ~ model + response,
  coef_map = c(
    "pedu_bi", "income_bi", "peduXincome",
    "pexpedu", "book", "art", "reloc",
    "acadcomm", "talentcls", "abstudy", "edusav",
    "(Intercept)"),
  gof_omit = "R2|BIC|RMSE",
  title = "Parenting and School Type",
  stars = TRUE,
  output = "markdown")

```

Table 6: Parenting and School Type

	(1)		(2)	
	star	priv	star	ordi
pedu_bi	0.735*** (0.085)	0.427*** (0.116)	0.308* (0.120)	-0.427*** (0.116)
income_bi	0.449*** (0.086)	0.807*** (0.089)	-0.359*** (0.107)	-0.807*** (0.089)
peduXincome	-0.057 (0.122)	-0.413** (0.151)	0.356* (0.151)	0.413** (0.151)
pexpedu	0.478*** (0.051)	0.689*** (0.072)	-0.212** (0.081)	-0.689*** (0.072)
book	0.100* (0.043)	0.040 (0.067)	0.060 (0.069)	-0.040 (0.067)
art	0.184* (0.072)	-0.150+ (0.084)	0.335*** (0.092)	0.150+ (0.084)
reloc	0.583*** (0.110)	0.154+ (0.090)	0.430*** (0.128)	-0.154+ (0.090)
acadcomm	-0.107* (0.051)	0.223*** (0.059)	-0.330*** (0.069)	-0.223*** (0.059)
talentcls	0.400***	0.648***	-0.248**	-0.648***

	(1)		(2)	
	star	priv	star	ordi
	(0.058)	(0.078)	(0.091)	(0.078)
abstudy	0.116	0.629***	-0.513***	-0.629***
	(0.086)	(0.107)	(0.111)	(0.107)
edusav	0.032	0.541***	-0.509***	-0.541***
	(0.047)	(0.066)	(0.072)	(0.066)
(Intercept)	-3.497***	-5.161***	1.664***	5.161***
	(0.306)	(0.409)	(0.473)	(0.409)
Num.Obs.	18465		18465	
AIC	26396.7		26396.7	
Std.Errors	by: sch_id		by: sch_id	
• p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001				

End of replication document

```
sessionInfo()
```

```
R version 4.5.3 (2026-03-11)
Platform: aarch64-apple-darwin20
Running under: macOS Tahoe 26.5.2
```

```
Matrix products: default
```

```
BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
```

```
locale:
```

```
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
```

```
time zone: America/New_York
```

```
tzcode source: internal
```

```
attached base packages:
```

```
[1] stats      graphics  grDevices  utils      datasets  methods    base
```

```
other attached packages:
```

```
[1] kableExtra_1.4.1  modelsummary_2.6.0 broom_1.0.13      lmtest_0.9-40
[5] zoo_1.9-1         sandwich_3.1-3     mlogit_2.0-0      nnet_7.3-21
```

[9] fixest_0.14.2	fastDummies_1.7.6	haven_2.5.5	lubridate_1.9.5
[13] forcats_1.0.1	stringr_1.6.0	dplyr_1.2.1	purrr_1.2.2
[17] readr_2.2.0	tidyr_1.3.2	tibble_3.3.1	ggplot2_4.0.3
[21] tidyverse_2.0.0			

loaded via a namespace (and not attached):

[1] gtable_0.3.6	bayestestR_0.19.0	xfun_0.61
[4] insight_1.5.4	CompQuadForm_1.4.4	lattice_0.23-1
[7] tzdb_0.5.0	numDeriv_2016.8-1.1	vctrs_0.7.3
[10] tools_4.5.3	Rdpack_2.6.6	generics_0.1.4
[13] datawizard_1.4.0	pkgconfig_2.0.3	dfidx_0.2-0
[16] Matrix_1.7-6	tinytable_0.19.0	checkmate_2.3.4
[19] data.table_1.18.6.1	RColorBrewer_1.1-3	S7_0.2.2
[22] stringmagic_1.3.0	lifecycle_1.0.5	compiler_4.5.3
[25] farver_2.1.2	textshaping_1.0.5	htmltools_0.5.9
[28] yaml_2.3.12	Formula_1.2-6	pillar_1.11.1
[31] nlme_3.1-171	tidyselect_1.2.1	digest_0.6.39
[34] performance_0.18.2	stringi_1.8.9	micsr_0.1-5
[37] labeling_0.4.3	splines_4.5.3	fastmap_1.2.0
[40] grid_4.5.3	cli_3.6.6	magrittr_2.0.5
[43] survival_3.8-12	withr_3.0.3	dreamerr_1.5.0
[46] scales_1.4.0	backports_1.5.1	timechange_0.4.0
[49] rmarkdown_2.32	otel_0.2.0	hms_1.1.4
[52] evaluate_1.0.5	knitr_1.52	parameters_0.29.3
[55] rbibutils_2.4.1	viridisLite_0.4.3	rlang_1.3.0
[58] Rcpp_1.1.2	glue_1.8.1	xml2_1.6.0
[61] svglite_2.2.2	rstudioapi_0.19.0	jsonlite_2.0.0
[64] R6_2.6.1	systemfonts_1.3.2	tables_0.9.35